



GYM & FITNESS CLUB MANAGEMENT SYSTEM

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Abstract

- A web-based system for managing gym members, memberships, trainers, payments, and attendance.
- Provides separate features for Admin, Member, and Trainer roles. Centralizes gym operations and improves accessibility, tracking, and management.

Problem Statement & Objectives

- Problem: Manual gym management makes it difficult to efficiently handle members, memberships, payments, attendance, and trainer activities.
- Objective 1: Provide secure Admin, Member, and Trainer access.
- Objective 2: Manage members, trainers, memberships, and workout plans.
- Objective 3: Automate payment and attendance management.
- Scope: Users: Admin, Member, Trainer Modules: Authentication, Membership, Payments, Attendance, Trainers, Workout & Diet Plans, Admin Management

Existing vs Proposed System

Feature	Existing System	Proposed System
Member Management	Manual Records	Digital management
Attendance	Manual entry	QR-based & Facial recognition
Membership	Paper/manual tracking	Digital membership
Workout plans	Manual communication	Centralized workout & diet plans
Limitation / Gap	Time-consuming & difficult to track	Automated and centralized system

Feasibility Study

- Technical: React.js, Node.js, Express.js and MySQL are available and suitable.
- Economic: Uses open-source technologies with low implementation cost.
- Operational: Simple web interface for Admin, Members and Trainers.
- Schedule: Can be developed and tested within the semester timeline.

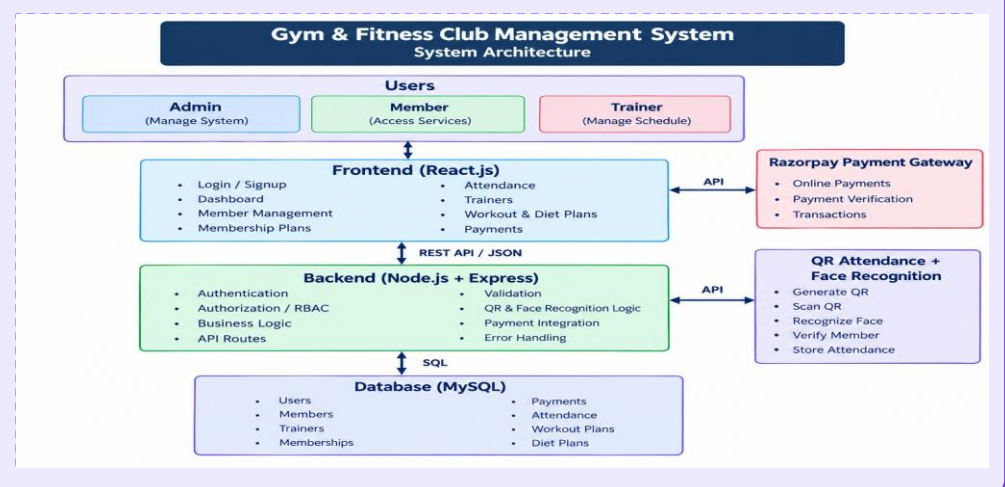
Results & Testing

- Successfully implemented role-based access for Admin, Member and Trainer.
- REST APIs connect the React frontend with the Node.js backend. MySQL stores and manages application data.

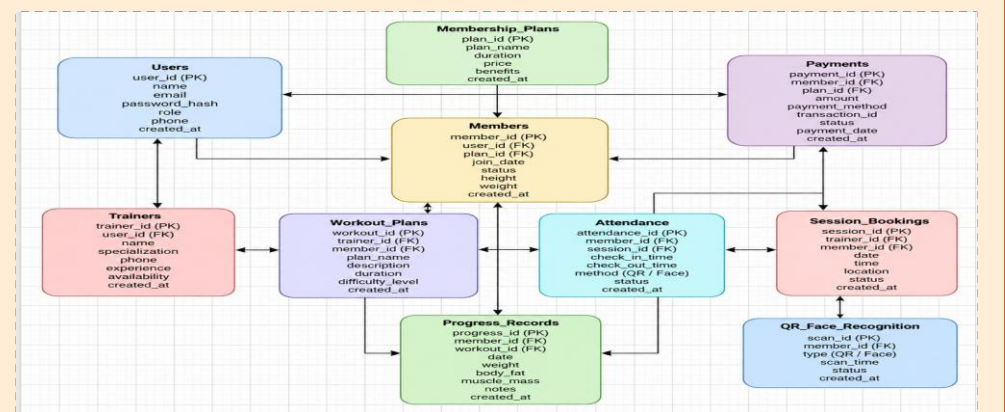
Tools & Technologies

- Front-End: React.js, HTML, CSS, JavaScript
- Back-End: Node.js, Express.js, REST APIs
- Database: MySQL
- Payment: Razorpay
- Attendance: QR Code
- Development: VS Code, Git/GitHub
- Testing: Postman, Browser Developer Tools

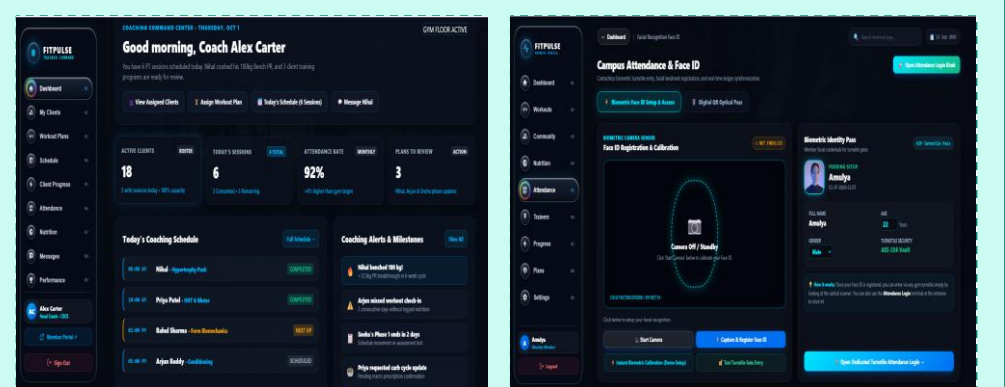
System Architecture



ER Diagram & Database Design



Implementation Screenshots



Conclusion

- Developed a centralized web-based Gym & Fitness Club Management System. Simplifies member, membership, trainer, payment and attendance management.
- Provides role-based access and digital gym services.

Future Scope

- Mobile application for members and trainers.
- AI-based personalized workout and diet recommendations.
- Advanced fitness progress analytics.

